INCH-POUND

A-A-50607(SH)
27 January 1994

COMMERCIAL ITEM DESCRIPTION

RESCUE SWIMMER DRY SUIT

1.0 ABSTRACT

This Commercial Item Description covers a dry suit to be used by US Navy surface rescue swimmers in all expected ranges of water temperature.

2.0 SALIENT CHARACTERISTICS

2.1 Design and construction. The rescue swimmer dry suit shall be a front entry (bib style or diagonal zipper) dry suit, black in color and made of a tri-laminate material (65 gr/m^2 inner and outer nylon with an inner layer of 100 gr/m^2 butyl rubber). The neck and wrist seals shall provide a watertight seal using heavy duty latex (a minimum thickness of 0.635 mm (0.025 in)) and shall be capable of being changed quickly. The neck dam shall be able to be replaced in five (5) minutes or less and the wrist cuffs shall be replaced in three (3) minutes or less. The quick-change neck and wrist seals shall interface directly with the suit material without any latex being between the attachment mechanism and the suit. The neck seal shall consist of a semi-rigid composite natural and synthetic rubber neck ring grommet bonded to suit, with a 6.35 mm (1/4 in) neoprene chaffing pad bonded to bottom side of neck ring. The neck seal shall fit over the neck ring and be secured with a stainless steel clamping band. The wrist seal shall consist of a semi-rigid composite natural and synthetic rubber wrist grommets bonded to suit. The wrist seals shall fit over the wrist rings and be secured with cuff retaining rings. The neck and wrist seals shall have lines around them to guide in trimming. The knee and buttock areas of each dry suit shall be made of reinforced material. Molded boots shall be used. The dry suit shall also consist of the following components:

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AMSC N/A FSC 4220

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2.1.1 Suspenders. Each dry suit shall be equipped with 38 mm (1-1/2 in) wide elastic, adjustable suspenders with 38 mm (1-1/2 in nylon "quick-release" buckles secured inside the suit. The suspenders shall be permanently attached to the suit at waist height in the rear and shall be crossed in the back and sewn. The attachment point of the buckles shall be in the front at waist height. The length of each suspender, prior to being stretched, from the attachment point in the rear to the base of the buckle shall be a minimum of 1.07 mm (42 in).

2.1.2 Exhaust valve. Each dry suit shall be equipped with a low profile, automatic, ambient exhaust valve mounted in the right forearm for the release of air trapped inside the suit. The breaking pressure of the exhaust valve shall be a maximum of 7.6 cm (3.0 in) of water gauge.

2.2 Seams. The dry suit seams shall be tested for breaking strength using method 5100 of FED-STD-191A, Federal Standard for Textile Test Methods where applicable as described in section 3.5. The dry suit seams (without reinforcement) shall have an initial and after ageing minimum breaking strength of 155.75 N (35 lbs) when tested as specified in section 3.5.

2.3 Sizing. The suits shall come in size large as sized by commercial standards with size 10 rubber boots for the feet. The boots shall be reinforced in the heel area, the back seam joint and the toe. Also, the boots shall have a neoprene sock inset with a thickness of 4 to 5 mm (.16 to .20 in) and a sole with a minimum thickness of 7.49 mm (0.30 in).

2.4 Labeling. The dry suits shall be labeled as follows: "US Navy Rescue Swimmer Dry Suit", manufacturer's name, contract number, date of manufacturer, and any special instructions.

2.5 Repair kit. A repair kit shall be furnished for each dry suit. The repair kit shall include the needed material and equipment for making minor repairs with instructions for care and repair of the suits. Also, one (1) extra neck seal and two (2) extra wrist cuffs with installation instructions shall be provided for each dry suit.

2.6 Product demonstration model. During the solicitation phase, interested firms shall provide the contracting agency with one product demonstration model of the item they propose to furnish for evaluation.

3.0 QUALITY ASSURANCE

3.1 Workmanship. The dry suits shall be uniform in quality and shall be free from irregularities or defects which adversely affect performance, reliability, or durability.

3.3 Examination of the end item. Examination of the assembled dry suit shall be made in accordance with table 1 prior to the leak test described in section 3.4.

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3.4 Leak testing. Prior to shipping, each dry suit shall be inflated to one (1) psi with the sleeves, neck, and exhaust valve plugged off, submerged, chest up, in a test tank and tested for leaks. The suit shall then be rotated 180 degrees and retested to ensure that all areas of the suit have been tested for leaks. The plugging of the sleeves, neck, and exhaust valve shall be done in a manner which will cause no temporary or permanent damage to the dry suit. One leak per suit is allowed and shall be repaired only once. If the suspect area shows further leakage after repair, the suit shall be rejected. Any suit containing more than one leak shall be rejected.

3.5 Seam testing. Using the test apparatus and procedures described in method 5100 of FED-STD-191, the seam samples shall be tested for breaking strength. The specimens shall be centered equidistant from the clamps. All pulls shall be made with the seam at right angles to the load direction. The clamps shall be one inch apart at the start of the test and the face of each jaw shall be not less than 2.54 cm by 5.08 cm (1.0 by 2.0 in). The seam test specimen shall be rectrangular, 5.08 cm (2.0 in) wide by not less than 10.16 cm (4.0 in) long and shall be prepared a minimum of 12 hours prior to testing. The seams shall be aged by being suspended vertically in an air-oven at 70 ± 1 degree Celsius (158 ± 2 degrees Fahrenheit) for seven days. At the conclusion of the ageing interval the specimens shall be cooled at room temperature on a flat surface and allowed to rest for 4 hours before determination of breaking strength. Two sets of five specimens each shall be tested: one set of five specimens not aged and one set of five specimens after ageing. Upon completion of the breaking strength test, the five results for each set (not aged and after ageing) shall be averaged. These results shall meet the criteria of section 2.2 for initial and after ageing minimum breaking strength.

4.0 PACKAGING, PACKING AND MARKING

4.1 Packaging, packing and marking. The dry suits shall be packaged in a watertight container such that the suit is not damaged by the neck ring during shipping. Each package shall contain one dry suit and one repair kit.

5.0 NOTES

5.1 Intended use. This Commercial Item Description covers a dry suit to keep rescue swimmers dry during rescue operations.

5.2 Material substitution. The material selected for use is a tri-laminate material composed of 65 gr/m^2 inner and outer nylon with an inner layer of 100 gr/m^2 butyl rubber. This material exhibits the mobility and buoyancy requirements of the rescue swimmers. Any substitution of material is prohibited.

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TABLE 1. Examination requirements.

Defects	Classification	
	Major	Minor
I. Material defects and workmanship damages		
a. Any hole	X	
b. Deep crease or fold line	X	
c. Cut or tear	X	
II. Basic material		
a. Any evidence of delamination	X	
b. Area of excess cement		X
c. Any brittle area which cracks when bent	X	
d. Any foreign material which when removed results in damage to suit	X	
e. Any foreign material which when removed does not result in damage to suit		X
f. Any blister or pin hole	X	
g. Any objectionable odor		X
h. Color not as specified	X	
i. The dry suit material is not the material specified	X	
III. Cleanness		
a. One or more spots or stains of a permanent nature		X
IV. Component and assembly		
a. Any component part omitted or any required operation omitted or improperly performed	X	
V. Seams		
a. Any open seam	X	
b. Seam not properly joined	X	
VI. Automatic exhaust valve		
a. Item does not work as intended	X	
b. Item not located on suit as specified	X	
c. Item not secured on suit properly	X	
d. Suit leaks water around valve	X	
VII. Wrist cuffs		
a. Omitted	X	
b. Wrist cuffs not secured to suit properly	X	
VIII. Neck seal		
a. Omitted	X	
b. Neck seal not secured to suit properly	X	
IX. Labeling		
a. The label is not legible or does not contain the required information		X

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